The Evolution of Transit Access to Jobs in American Cities

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Draft: November 10, 2026

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Abstract

Households without a car are largely limited to the jobs they can reach by public transportation. We measure job access by transit as the jobs reachable within 45 minutes on the scheduled transit network. No consistent sub-annual series of this measure has covered many U.S. metropolitan areas over a decade. We construct a quarterly neighborhood panel for 52 metropolitan areas over 2014–2025, routed from archived schedules with one fixed method. Access depends on the transit schedule, the number of jobs, and where they are, and we separate the three. The typical neighborhood's job access by transit rose about 15%, falling below its 2014 level in 2020 and ending above its pre-pandemic peak. In metros with clean archives, job growth alone would have produced more than the entire rise. The schedule's own contribution was roughly nothing before 2020 and negative after, a level we compare with agency-reported service data, and jobs relocating toward transit-served places offset most of that loss. The growth went to richer, car-owning, peripheral neighborhoods. The poorest quartile and majority-Black neighborhoods gained little or lost access, depending on whether operators added to the archive during the decade count as new service. Within cities, at the same distance from the same core, growth accrued to richer neighborhoods, overwhelmingly through where employment located. On the corrected series, weighted by car-free households, the rise is +2.5%, against +15% for the average resident. The paper is descriptive; we make no causal claims about employment.

Keywords: Job accessibility; public transit; GTFS archives; spatial mismatch; transport equity

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1 Introduction

About 8% of U.S. households have no vehicle available (U.S. Census Bureau, 2023). Not driving is more common among lower-income and Black adults. In a 2024 survey, 19% of lower-income adults and 21% of Black adults said they rarely or never drive, compared with 3% of upper-income adults and 7% of White adults (Pew Research Center, 2024). For workers in these households, the jobs they can reach by transit make up much of the set of jobs they can take. Research on spatial mismatch has long linked access to jobs with employment for workers without cars.

Transit agencies decide where and how often buses and trains run, and they revise their schedules several times a year. They report the service they operate and the trips riders take to the National Transit Database (NTD). Neither number shows how many jobs residents can reach. Regional transportation plans now commonly list access to jobs among their goals, but few plans measure it. Federal civil rights guidance also requires large agencies to test whether major service changes fall disproportionately on minority and low-income populations. Each of these tasks depends on knowing how many jobs residents can reach by transit, and how that number changes when service changes. Section 2 reviews this research and practice.

Between 2014 and 2025, both the transit network and the location of jobs changed. In 2020 agencies cut service as ridership fell. In 22 U.S. cities, those cuts reduced transit access to food and health care most in neighborhoods with high poverty and many households without vehicles (Kar et al., 2022). Over the same decade, metropolitan areas added jobs, and the location of those jobs shifted within each metro. The number of jobs a neighborhood can reach by transit depends on the network and on the number and location of jobs. Service statistics alone therefore cannot show whether residents ended the decade able to reach more jobs by transit or fewer. This paper asks how job access by transit changed across U.S. neighborhoods from 2014 to 2025. It also asks how much of the change came from transit service and how much from jobs, and which neighborhoods gained or lost.

Prior work has established that job access by transit is low in most U.S. metropolitan areas and far below job access by car. The *Access Across America* series reports annual levels for metropolitan areas (Owen et al., 2025). The *Modal Access Gap* program maps the transit/car gap by block group (Maharjan et al., 2024). The Transit Equity Dashboard tracks block-group access in seven regions from 2020 onward (Klumpenhouwer et al., 2021), and a recent study follows New York City block groups across four snapshots (Zhang et al., 2026). None of these sources measures neighborhood job access by transit across many metropolitan areas, more often than once a year, with one method held fixed for a decade.

Without such a series, a change in measured access cannot be distinguished from a change in method or data. The parts of a change that are due to the transit network and to the location of jobs also cannot be separated.

We construct a quarterly panel of job access by transit for every census block group in 52 U.S. metropolitan areas and in New York, from 2014 to 2025. We refer to block groups as neighborhoods. For each neighborhood and quarter, we count the jobs a resident can reach within 45 minutes by transit and walking, departing in the morning peak. Travel times are computed from archived transit schedules in the General Transit Feed Specification (GTFS) format, with one routing method held fixed across all metros and quarters. Job counts come from the Census Bureau's Longitudinal Employer–Household Dynamics (LEHD) Origin–Destination Employment Statistics (LODES). We analyze New York separately (Section 6), because its job access by transit is about ten times that of the other metros and the way its access varies with neighborhood income and race differs from theirs.

We measure change by comparing each neighborhood with itself over time. Because the schedule archive added transit operators over the decade, we do not count an operator's first appearance in the archive as new service. We then split each neighborhood's change into three parts, using the 19 metros whose archived schedules have no quarter with a data problem. The first holds jobs fixed and lets only the transit schedule change. The second is growth in the metro's total number of jobs, and the third is the movement of jobs within the metro. We also report how many times more jobs a resident can reach by car than by transit. Outside New York the multiple is an order of magnitude or more. Our car travel times assume free-flowing traffic, so this multiple overstates the car advantage. The paper is descriptive. It measures access to jobs, not employment.

We find three things. First, the typical neighborhood's job access by transit rose about 15% from 2014 to 2025, after falling below its 2014 level in 2020. The 15% is the population-weighted average of each neighborhood's own change. Second, the rise came from jobs. In those 19 metros, growth in the number of jobs alone would have produced more than the entire rise. The transit schedule added almost nothing before 2020 and reduced access after 2020, and jobs moving toward places served by transit offset most of that reduction. Third, the gains went mainly to higher-income, car-owning neighborhoods at the edges of metropolitan areas. Weighted by households without a car, access rose 2.5%, compared with 15% for the average resident. For practice, these results show that service statistics and job access can move in opposite directions, and that averages over all residents can overstate what car-free households gained.

The paper proceeds as follows. Section 2 reviews the related literature and planning practice. Section 3 describes the data and the construction of the panel. Section 4 defines

the access measure and the methods for measuring change. Section 5 presents the results. Section 6 reports the results for New York. Section 7 discusses the limitations and the implications for practice, and concludes.

2 Related Literature

Accessibility measures the opportunities that can be reached from a location within a given travel budget, a concept dating to Hansen (1959). Subsequent work has developed cumulative-opportunity, gravity-based, and utility-based measures and has clarified the strengths and limitations of each approach (Handy and Niemeier, 1997; Geurs and van Wee, 2004). In practice, cumulative-opportunity and gravity-based measures can give nearly the same results. In Montreal, they are correlated above 0.9 when the cumulative threshold is set at the mean commute time (Santana Palacios and El-Geneidy, 2022). Accessibility fluctuates within the day, though a constant peak-hour measure tracks relative accessibility across neighborhoods closely (Boisjoly and El-Geneidy, 2016). A further methodological thread decomposes a change in accessibility into the part attributable to the transport network and the part attributable to the population or land-use (here, job) distribution (Stepniak and Rosik, 2018; Rosik et al., 2025). A related literature separates the same two contributions across design scenarios (Lahoorpoor et al., 2022). Applying these decompositions to observed changes over time requires accessibility measures computed consistently across years. Such measurement is feasible with general-purpose routing over open schedule data (GTFS) and street networks (Conway et al., 2018; Pereira et al., 2021). The closest precedent for historical measurement reconstructs a retrospective timetable from vehicle-location data (Wessel et al., 2017).

While this literature establishes how accessibility is measured, the more important question for this paper is what those measures reveal about U.S. transit systems. Applied to U.S. transit, these measures have produced a body of direct evidence on how little of a metropolitan job market a rider can reach. Tomer et al. (2011) provide a 100-metro cross-section, and the *Access Across America* series (Owen et al., 2016), building on continuous-accessibility methods (Owen and Levinson, 2015), reports transit accessibility annually, with a parallel automobile series. Its recent editions report year-over-year comparisons (Owen and Murphy, 2020) and describe a decade of consistently calculated data (Owen et al., 2025). Because each edition pairs that year's schedules with that year's employment data, its year-over-year changes combine service change with economic change. The Transit Equity Dashboard (Klumpenhouwer et al., 2021) tracks block-group transit access and its demographic incidence in seven large U.S. regions from February 2020 forward. Closest to our object is the

Modal Access Gap program: transit versus car job access computed at the block-group level across the largest U.S. metropolitan areas, with an explicit focus on carless and underserved populations (Maharjan et al., 2024). The same group measures transit against walking in the 50 largest metros, reading the gap as an effectiveness indicator (Ermagun and Witlox, 2024). This work establishes the cross-sectional gap and its spatial structure, including the finding that carless households tend to live where the gap is narrower (Maharjan et al., 2024); its temporal dimension is travel time rather than calendar time (Janatabadi et al., 2023).

What gives these magnitudes their economic weight is the spatial mismatch hypothesis (Kain, 1968), which holds that physical disconnection from employment depresses labor-market outcomes for disadvantaged workers. The hypothesis and its mechanisms have been reviewed extensively (Ihlanfeldt and Sjoquist, 1998; Gobillon et al., 2007). A strand of this work stresses that mismatch is as much modal as spatial, because a worker without a car faces a far smaller opportunity set than a driver from the same location (Grengs, 2010). Studies of transit itself find that better access to public transit is associated with higher labor-force participation in Portland and Atlanta (Sanchez, 1999). When damage from Hurricane Sandy reduced subway service to parts of New York, unemployment rose in the affected neighborhoods, most among residents who depend on transit (Tyndall, 2017). A systematic review of 93 studies finds a positive association between transport and employment. The association is clearest for car ownership, and the evidence for public transit is thinner (Bastiaanssen et al., 2020). Kain's hypothesis also has a job side. Employment in U.S. metros has decentralized for decades (Glaeser and Kahn, 2001), with job share shifting away from the urban core in nearly every large metro through the 2000s (Kneebone, 2009). Kain identified that movement as a cause of the mismatch. Whether decentralization has been pulling jobs toward or away from the places transit serves is an empirical question about the joint evolution of the two.

How access is distributed across people and places has been studied on both empirical and normative grounds; the normative case is reviewed by Pereira et al. (2017). Foth et al. (2013) track transit access against neighborhood social need in Toronto across two census years, and Yeganeh et al. (2018) compare access by income and race across U.S. metropolitan areas in a single cross-section. Most closely, Zhang et al. (2026) track block-group transit access to jobs and services across four snapshots for New York City, decomposing a widening racial access gap into transit-network and residential-sorting components.

Accessibility also has a growing place in planning practice. A review of 32 metropolitan transportation plans from North America, Europe, Australia and Asia finds that most include accessibility among their goals, most often as access to jobs, but few use accessibility-based indicators to guide decisions (Boisjoly and El-Geneidy, 2017). A content analysis of 42

U.S. regional transportation plans finds that most include accessibility-related goals, but few define the term or use accessibility-oriented performance measures (Proffitt et al., 2019). Equity objectives in North American plans are often stated without measures that can assess their achievement for different population groups (Manaugh et al., 2015). For transit service, federal civil rights guidance requires agencies that operate 50 or more fixed-route vehicles in peak service, in urbanized areas of 200,000 or more people, to conduct service equity analyses of major service changes (Federal Transit Administration, 2012). The outcome of such an analysis can depend on whether it uses census or onboard-survey data (Karner and Golub, 2015). Agencies report the service they operate and the trips riders take to the National Transit Database each month (Federal Transit Administration, 2026). In 2020, agencies cut frequencies in many areas and raised them in others, and the cuts appear to have fallen disproportionately on lower-income and more vulnerable areas of 40 North American cities (DeWeese et al., 2020). The pandemic's fall in transit demand is documented across U.S. systems (Liu et al., 2020).

Taken together, these literatures establish the level of U.S. job access by transit, the cross-sectional shape of its modal gap, its distribution at a point in time, and its use as a planning goal, and they supply the methods for decomposing a change in access. The object that remains open is job access by transit observed in calendar time: at the block group, on a single method held fixed, at sub-annual resolution, across many metros. That is the panel this paper builds.

3 Data

We assemble a neighborhood-level panel of job access by transit and by car for 52 U.S. metropolitan areas outside New York, plus New York itself, measured quarterly from 2014 through 2025. It combines four sources: an archive of transit schedules, the location of jobs, the street network, and demographic context. Assembling consistently routable schedules at sub-annual resolution, on the same dates in every metro across more than a decade, has not been done before.

3.1 Data sources

Transit service is described by GTFS feeds. GTFS defines a common format for public transportation schedules and the geographic information that accompanies them. An agency publishes its routes, its trips, the location of every stop, and the time a vehicle is scheduled to serve each one. The specification exists so that this data can be consumed by software, which

is what puts transit directions into applications like Google Maps, and nearly every transit agency in the country adheres to it. There is, however, no national repository of feeds. Each agency keeps its own schedules, and superseded versions are ordinarily discarded, so a feed downloaded today describes only today's service. What makes a historical panel possible is that community archives capture dated snapshots and keep them. We assemble ours from two such archives, Transitland and the Mobility Database, supplemented for Philadelphia's recent years by the archive published by its transit agency, the Southeastern Pennsylvania Transportation Authority.

Job locations come from the LODES Workplace Area Characteristics, which report job counts at the workplace Census block annually; we aggregate them to the block group, so that both ends of a trip are measured on the same geography. We count all jobs throughout, using the total job count (C000) of LODES job type JT00, which also counts workers' secondary jobs. LODES also separates jobs by earnings band, but its low-wage band is a fixed nominal threshold that holds fewer jobs each year as wages grow, so no low-wage access measure is reported. Version 8 of LODES reports every year on 2020 census blocks, so the destination geography is stable across the panel. LODES counts the jobs held at the beginning of the second quarter of each year, in April, and we apply each year's counts to all four quarters of that calendar year. We use LODES through 2023, the latest year available when the panel was built, so the 2024 and 2025 quarters use 2023 job counts. Alaska's files end in 2016 and Michigan's in 2021, so Anchorage and Grand Rapids use their last available year after those dates. Because the jobs data are annual, sub-annual movement within any year is driven entirely by changes in the transit schedules.

Door-to-door routing uses the OpenStreetMap (OSM) street and path network, which supplies the walking legs of transit trips and the driving times for cars. The car alternative is free-flow using posted speeds with no congestion, parking, or access-and-egress penalty, which overstates real driving access. One present-day OSM extract is used for every quarter, so the street network is held fixed and changes over time come from the transit schedules and the jobs.

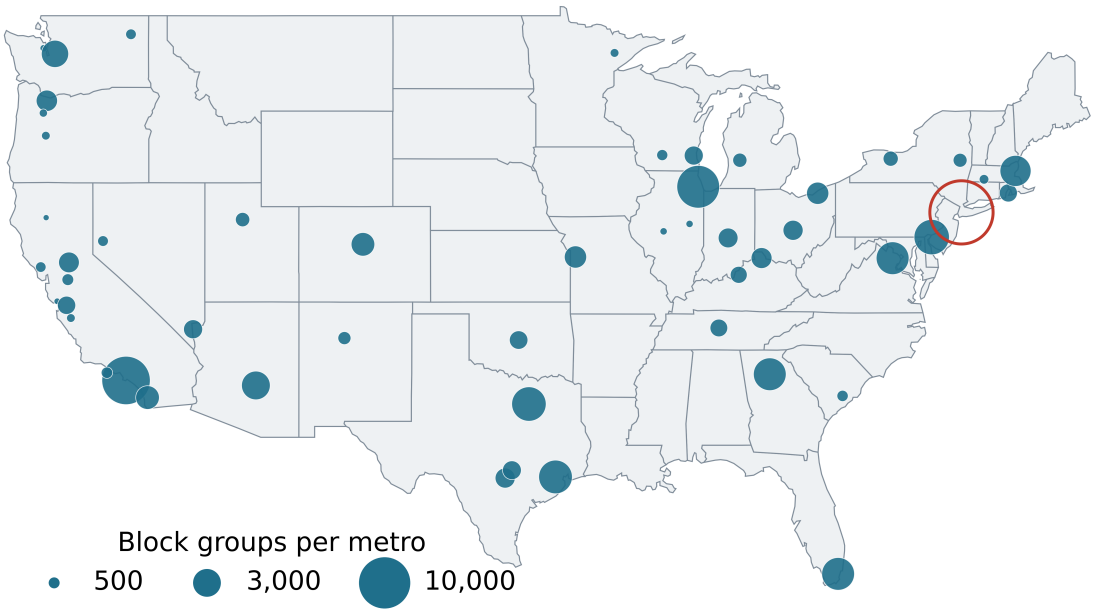
Neighborhood population, income, race, and vehicle availability come from the 2018–2022 American Community Survey (ACS), held fixed across quarters. The population counts are the weights of every population-weighted statistic. We use them to characterize who has car access and to identify transit-reliant neighborhoods, those with high shares of car-free households (households with no vehicle available) and high poverty shares. A neighborhood's poverty share is the share of its residents whose income is below twice the poverty level. Both conditions are measured as neighborhood shares, not household by household, so the car-free households in a transit-reliant neighborhood need not be the poor ones. Each neighborhood's

distance to the urban core is its distance to the nearest employment center. A metro's employment centers are the block groups with the highest job density that together hold a third of its 2023 jobs.

3.2 Panel construction

Agencies republish their feeds at every service change, several times a year in the recent era, and these versions are archived, so sub-annual sampling resolves service changes, above all the COVID-19 service cuts and recovery, that an annual snapshot would miss. Quarterly sampling is feasible only in the recent era. Routed quarters exist back to 2013, but the archive there is thin, so the analysis window, and every exhibit, runs 2014–2025.

Figure 1: The 52 panel metropolitan areas outside New York



Note: The 52 U.S. metropolitan areas outside New York analyzed at quarterly resolution, 2014–2025; point area is proportional to the number of block groups. New York, analyzed separately in Section 6, is the hollow marker. Anchorage, also in the panel, lies outside the map.

For each operator serving a metro we take the feed in effect on the second Wednesday of January, April, July, and October of each year (a representative mid-week service day), combine all the operators that serve the metro into a single routable network, and route that schedule. A metro-quarter is thus the service scheduled on that date. Quarter-of-year effects are small. Averaged over the decade, each quarter of the year lies within about a log-point of its year's mean (a log-point is 100 times a difference in natural logarithms, close to one percent for small changes). The choice of the second Wednesday therefore does not drive

the results. We do not substitute feeds across dates. A metro-quarter includes an operator only when its feed is strictly contemporaneous, meaning its service period spans the sample date; an operator without one is dropped from that quarter rather than backfilled with a stale feed.

Coverage in the archive is uneven (sparse in the early years, densest in 2016–2019, and thinner through a 2020–2023 trough), so the analysis uses a near-balanced panel: the 52 metropolitan areas outside New York present in at least 36 of the 48 quarters of 2014–2025. The set of metros is thus nearly the same in every quarter. We refer to it as the panel, and Table A.1 gives each metro's coverage. Each metro is a metropolitan statistical area, and the panel covers all of its block groups. Figure 1 maps the 52, with marker area proportional to each metro's number of block groups. They range from large metros such as Chicago and Los Angeles to small ones such as Champaign–Urbana and Redding. New York is analyzed separately (Section 6). Its job access by transit is about ten times that of the other metros, so it would dominate any population-weighted average, and several of its differences in access across neighborhood groups run in the opposite direction. The panel is the sample of every level, trajectory, and distributional exhibit. Exhibits that compare the start and end of the decade narrow it to the 45 metros observed in both 2014–15 and 2024–25, and the decomposition uses a stricter sample (Section 4.3).

3.3 Data quality and representativeness

Archived feeds can misrepresent the schedule in effect. We scan each metro's quarterly series for an operator missing from a quarter between quarters in which it is present (a partial outage, which would understate that quarter's access), for a quarter with implausibly little service, and for access that spikes or stays depressed because a feed was included twice or operators went missing from the archive. We flag the affected metro-quarters rather than dropping them. Flagged cells are excluded from every access estimate but still count as present for coverage. An operator whose feeds were never archived in the early years is invisible to these scans, which detect changes rather than absences. This archive adoption is handled by the archive-entry correction (Section 4.2), which does not count an operator's entry into the archive as growth. Because internal scans cannot catch everything, we also compare the archive with the service agencies report to the National Transit Database, and that comparison flagged 69 metro-quarters that had passed every internal check. Table A.2 lists the flag classes, their detection rules and counts. Including the flagged cells would put the decade's rise in job access by transit at +17.7% rather than the +15.1% reported

in Section 5.1. Chicago's archive was degraded after 2020, and we recovered those quarters' schedules from additional versions in the Transitland archive. Finally, GTFS encodes scheduled rather than delivered service (Section 7).

Table 1 reports descriptive statistics for the panel, for the access measures pooled over block groups and quarters (Panel A) and for the neighborhood characteristics (Panel B). Job access by transit is far more skewed than job access by car. Within 45 minutes the mean block group reaches about 44,000 jobs by transit and the median about 6,800, against about 1.5 million and 1.2 million by car.

Table 1: Descriptive statistics, 52-metro panel outside New York

	Mean	SD	p10	p50	p90
A. Job access (block group × quarter)					
Transit jobs ≤ 45 min	44,214	115,756	276	6,768	107,221
Car jobs ≤ 45 min	1,536,484	1,204,558	237,127	1,181,697	3,163,993
Transit/car ratio R	0.034	0.120	0.0005	0.0058	0.100
B. Neighborhood characteristics (block group)					
Population	1,500	800	686	1,374	2,455
Median household income ($000)	96.7	49.1	43.5	86.8	164.4
Poverty share (%)	27.0	20.4	4.8	22.0	57.2
Car-free households (%)	7.7	11.7	0.0	3.2	21.9
Non-Hispanic White (%)	52.1	30.7	6.0	56.5	90.8
Non-Hispanic Black (%)	12.9	21.8	0.0	3.2	42.1
Hispanic (%)	23.0	25.5	0.2	12.9	65.1
Non-Hispanic Asian (%)	7.3	12.1	0.0	2.3	21.4
Distance to urban core (km)	7.2	11.0	0.6	3.4	18.3

Note: Panel A is unweighted, pooled over block-group × metro × quarter (flagged cells excluded), 45-minute budget, all jobs; R is the transit/car ratio, with free-flow car times (Section 4.1). Panel B is per block group ($n \approx 85{,}700$), from ACS 2018–2022 except the distance to the urban core. New York excluded.

The panel over-represents larger, transit-intensive metropolitan areas, because inclusion depends on historical GTFS availability rather than random sampling. The 52 cover about 45% of total U.S. metropolitan population, including 30 of the 50 largest metros and 40 of the largest hundred, and are more transit-oriented, though no more car-free, than the typical metro (Table A.3). They miss several large metros whose archives do not span the window: San Francisco–Oakland, Minneapolis–St. Paul, Detroit, and Baltimore. Washington is included, but the feed of its largest operator, the Washington Metropolitan Area Transit Authority (WMATA), is unarchived, so its access is understated throughout. We read the results as describing job access by transit in the covered metros rather than in the average U.S. metro.

4 Measuring Transit Access to Jobs

4.1 The accessibility measure

The unit of the panel is the block group × metro × quarter. For each metro-quarter we build a routable network from the metro's OpenStreetMap extract and the operators' strictly contemporaneous GTFS feeds and compute door-to-door travel times between every pair of the metro's 2020 block-group population centroids. The centroids serve as both origins and destinations, so jobs outside the metro are not counted.

Travel times are computed with R5, an open-source routing engine (Conway et al., 2018). For transit we take the median door-to-door time over departures in the 7:00–9:00 morning peak, so that a value reflects typical rather than best-case timing and does not depend on a single departure time. Trips longer than 90 minutes are treated as out of reach, well beyond the longest time budget in the paper (60 minutes). Walking uses R5's default speed with no separate distance cap, and all other routing parameters are R5's defaults, held fixed across every metro and quarter. Because routing runs centroid to centroid, travel times are coarsened in large peripheral block groups. The centroids are fixed, so this error is the same in every quarter. It does not cancel entirely from a block group's change, because a fixed error can still decide whether a destination falls inside the time budget when the schedule changes.

From the resulting travel-time matrix we form cumulative access: the count of jobs at destinations reachable within a fixed time budget. The cumulative job access by transit of origin block group i in metro-quarter q at budget τ is

$$A_{i,q}^{(\tau)} \;=\; \sum_{j} \text{jobs}_{j,y(q)} \cdot \mathbf{1}\Big[\, t_{ij}^{\text{transit}}(q) \le \tau \,\Big], \qquad \tau \in \{15, 30, 45, 60\} \text{ min},$$

where $t_{ij}^{\text{transit}}(q)$ is the transit travel time on the schedule operating in quarter q and $\text{jobs}_{j,y(q)}$ is all jobs (LODES C000) at destination j in the annual LODES vintage $y(q)$ of that quarter. We report the 45-minute budget as primary.

Figure 2 maps job access by transit for one mid-size panel metro. Access peaks downtown and falls by roughly two orders of magnitude toward the periphery. We summarize it population-weighted (the access of the average person, the convention in the cumulative-opportunity literature, Owen and Levinson, 2015) and report the median block group as a companion. Access is sharply right-skewed within a metro, near zero across the low-service periphery and high in the service-bearing core; in the metro mapped here the median block group reaches about 5,400 jobs.

Figure 2: Job access by transit per block group: Cincinnati, 2023Q4



Note: Jobs reachable within 45 minutes by transit, per block group, Cincinnati 2023Q4 (log scale, central 30 km).

Cumulative access is the most transparent and most widely reported construct. Two alternatives drop the hard cutoff: gravity measures, which count every job but discount it smoothly by travel time (here a negative exponential with a 45-minute half-life), and utility-based (logsum) measures, which summarize the expected value of choosing among all reachable destinations. With each decay parameter set at 45 minutes, both are about 0.9 correlated with cumulative access across block groups (Table A.5), as Santana Palacios and El-Geneidy (2022) find for gravity measures. These correlations compare levels in a single quarter. The rest of the paper uses the cumulative measure. Access climbs steeply with the time budget, roughly forty-five-fold between 15 and 60 minutes (Table A.4), but the overall series at each budget move in parallel across the decade, so the shape of the trajectory does not depend on the 45-minute choice. The group comparisons in Section 5.6 vary in size with the budget.

We check the measure against two external sources. Across the 390 metro-years that overlap the published *Access Across America* series, our metro levels agree with theirs at log correlations of 0.88–0.95 (Fig. A.5). Our levels sit at a stable fraction of the published values. The median ratio is 0.57 at 30 minutes and 0.64 at 60 minutes. The middle half of metro-years lies between 0.47 and 0.64 at 30 minutes and between 0.54 and 0.72 at 60 minutes. The fraction is below one because our metropolitan areas include a low-access periphery that the series' urban areas leave out, and because the two series weight neighborhoods differently. Ratios below 0.3 occur in 4% of metro-years at 30 minutes and 7% at 60 minutes, mostly in Washington, whose largest operator's feed is unarchived, and in Philadelphia and Miami. The measure also predicts behavior. Metros with more jobs reachable by transit have more workers who commute by transit ($r = 0.77$ outside New York; Fig. A.6).

Car access $A_{i,q}^{(\tau),\text{car}}$ is the same count with car travel times in place of transit times. These come from the same routing engine and the same fixed street network as the transit times, so the two modes differ only in how a traveler moves. The transit/car ratio $R_{i,q} = A_{i,q}^{(\tau),\text{transit}} \big/ A_{i,q}^{(\tau),\text{car}}$ at the same origin ($R = 0.01$ is a hundredfold car advantage) enters only as a secondary comparison, reported with the levels in Section 5.5. Because the car baseline is free-flow, the car advantage this ratio implies is an upper bound. We do not follow the ratio over time, because with the car network held fixed, its change mixes the change in job access by transit with the change in jobs within reach by car.

4.2 Measuring change

The trajectory is the change in job access by transit in each quarter relative to the 2014 annual average. It compares each block group only with itself, so the large differences in access between neighborhoods do not move it when some metros are missing from a quarter. Formally, it is the set of quarter coefficients from a population-weighted regression of log access on quarter indicators and block-group fixed effects. Log access is the natural logarithm of one plus the number of jobs reachable, which keeps block groups that reach no jobs in the sample. The paper refers to this trajectory as the change for the typical neighborhood or, since block groups are weighted by population, for the average resident.

The archive also grew over the decade. In every panel metro, at least one operator's schedules first appear in the archive after the metro's first year. Counted directly, each such entry would appear as new transit service. The archive-entry correction therefore counts change only within each period between operator entries into the metro's archive. In the quarter an operator enters, the jump in access is not counted as growth, and the comparison continues from the new level. This is the splice familiar from chained price indices, which

link an index across a change in its basket of goods. In the regression, the block-group fixed effect becomes one fixed effect for each block group and each such period.

The correction drops no observations. Every quarter stays in the panel, and only the step at an operator's entry is left out of the trend. Where an operator began running when it entered the archive, the correction also leaves that new service out of the trend, which understates growth in the neighborhoods it serves. We checked the larger entries against agency-reported service where that was possible. Each was the archive beginning to collect an existing operator's feed, except in Columbus, where the entry is new service. Treating the five cases in which an operator leaves the archive as real service changes moves the 2025 estimate by less than a tenth of a log-point. Figure A.1 shows the uncorrected and corrected series for Miami. Across the panel, the uncorrected index ends 2025 at about +22 log-points (+25%), against +14 for the corrected index (Fig. A.2). The difference of about eight log-points is mostly archive adoption rather than new service.

The within-city comparisons and the metro ranking in Section 5 use a single decade change for each block group instead of a quarterly path. It is the mean of log access in 2024–25 minus its mean in 2014–15, with each step at an operator's entry removed.

Confidence intervals throughout are clustered at the block group, which absorbs the serial correlation of a neighborhood with itself over time. By the design-based logic of Abadie et al. (2023) this is the right uncertainty for describing the panel we observe, since we hold essentially the entire population of its block groups. For claims that read as a common process across metros, we also report metro-clustered intervals (Table A.6) and scope the text to whichever the evidence supports.

4.3 The decomposition

The decomposition separates each block group's change in log access into a network part and a jobs part. Conceptually, we ask what access would have looked like if only transit had changed, and separately if only jobs had changed. Holding the 2014 job map fixed and letting only the network change isolates the network's own contribution; holding the 2014 network fixed and letting only jobs move isolates the job side. Measured this way, the two parts do not quite add up to the observed change, because the network and the job map also changed together. We therefore average each part over both choices of which year is held fixed, 2014 or the later year, and the averaged parts sum exactly to the observed change in every block group. The split into transport and land-use parts follows the decomposition literature (Rosik et al., 2025), and the averaging is the symmetric allocation of the Shapley construction (Shorrocks, 2013). The jobs part splits exactly in two. Metro-wide job growth is

the growth of the metro's job total, because growth in the same proportion everywhere raises every block group's access by that proportion. Job relocation is the remainder, employment shifting toward or away from the places transit serves. The decomposition is annual, because LODES reports jobs once a year, and its components are measured against the 2014 annual average, because the archive's early quarters are thin. It is also corrected for archive entry. An operator's entry changes the archived network, so in each block group the step in access at the entry is subtracted from both the observed change and the network part. The three parts then still sum exactly to the observed change.

The decomposition is estimated on the clean-archive core: of the panel's 52 metros, the 35 whose block groups are observed unflagged in every year, and among those the 19 with no flagged quarter at all across the decade (28,147 block groups). Metros with any flagged quarter are excluded whole, because their component paths track changes in archive quality rather than in service. Their measured networks appear to improve during the pandemic. The exclusion protects the component paths. The 2025 endpoints change by only a fraction of a log-point without it, and the jobs-side components are nearly identical on the full every-year sample (Fig. A.4). That comparison of 2025 endpoints holds the sample fixed and varies only the flag treatment.

5 The Evolution of Job Access by Transit

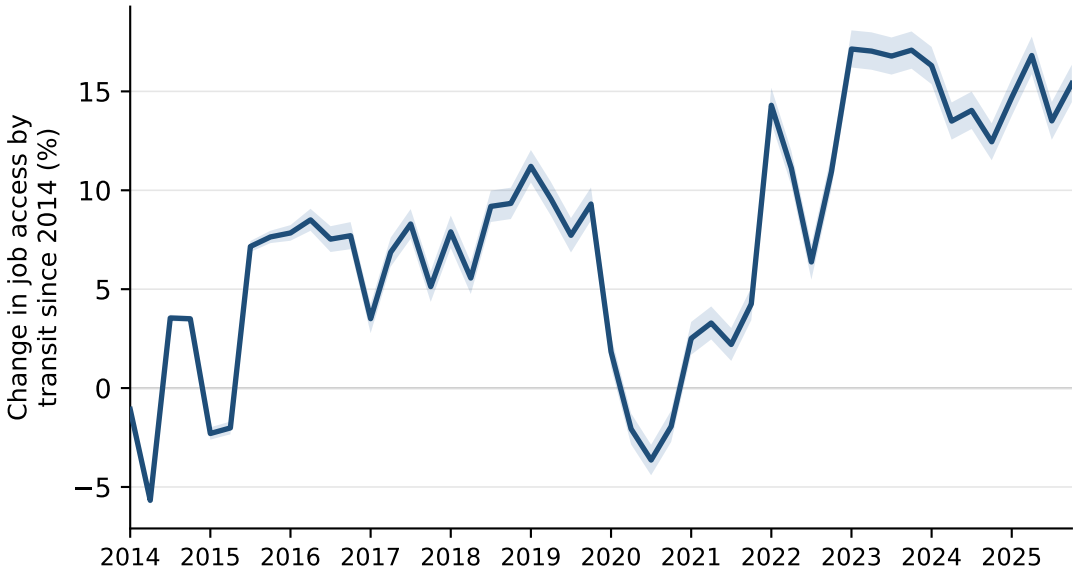
Unless noted, results are on the panel, the 52 metros outside New York (Section 3.2), population-weighted, at the 45-minute budget for all jobs; the decomposition alone uses the stricter clean-archive core. Trajectories compare each block group with itself and are corrected for archive entry (Section 4.2), and confidence bands are clustered at the block group. New York is analyzed separately in Section 6. Section 5.1 shows the trajectory, Section 5.2 decomposes it into network and jobs components, and Section 5.3 covers the pandemic. Section 5.4 compares metros. Sections 5.5 to 5.7 cover the distribution within cities, asking who has access, who gained it, and how car-free households fared.

5.1 Change in accessibility over time

Figure 3 shows the quarterly trajectory on the 52-metro panel, the change in the typical neighborhood's job access by transit since the 2014 average, with its 95% confidence band. The typical neighborhood's job access by transit rose about +9% by late 2019 (+8.9 log-points $[8.1, 9.7]$) and fell below its 2014 level during 2020 (-3.7 $[-4.5, -2.9]$ in 2020Q3). The fall began in 2020Q1, before the pandemic service cuts, with a drop of 7.1 log-points

from 2019Q4. That quarter is the first to use the 2020 job counts, which describe jobs held at the beginning of April 2020 (Section 3.1), so part of the 2020 fall is in the job counts. It recovered to about +15% by 2025 (+14.1 log-points [13.3, 14.8]; +15.1% in ratio terms), above its pre-pandemic peak. The recovery was uneven. The index first passed its 2019 level in 2022 and has stayed between about 12% and 17% since 2023. These intervals describe the panel; as a statement about a common national process the net decade change holds up, with a metro-clustered interval excluding zero (about [+6, +24] around the metro-clustered specification's own estimate of +14.9; Table A.6), though metros vary widely around it, the dispersion Section 5.4 documents.

Figure 3: Change in job access by transit relative to 2014, quarterly, 2014–2025



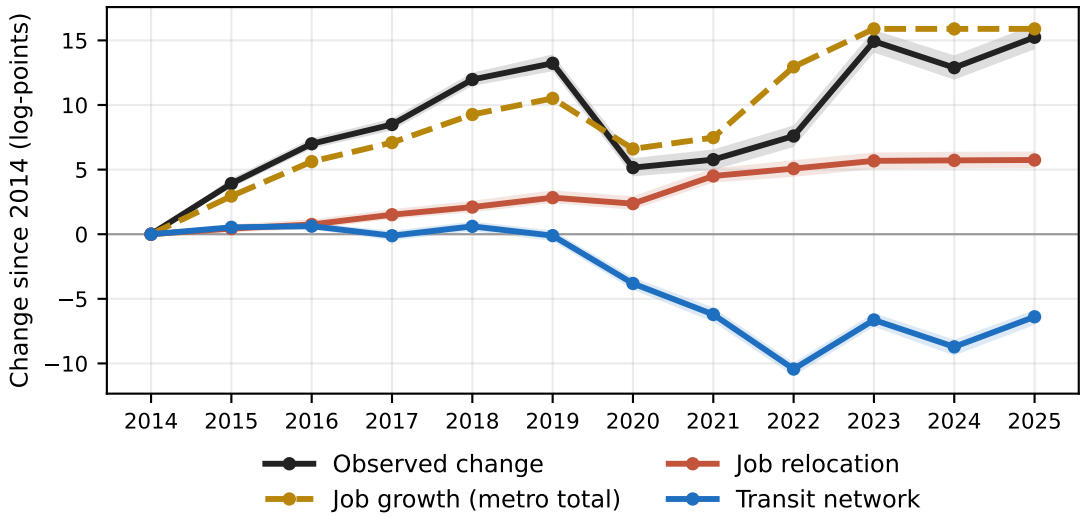
Note: Within-block-group fixed-effects change in jobs reachable within 45 minutes by transit, corrected for archive entry, relative to the 2014 average, population-weighted, 52 metros outside New York. Shading is the 95% block-group-clustered band.

5.2 What drove the rise: network or jobs?

The more informative question is what drove the rise. Did job access by transit improve because the network expanded, or because the jobs moved toward it? The decomposition of Section 4.3 separates the two.

Figure 4 shows the answer, which is the paper's central finding. It plots the observed change in log access since the 2014 average on the clean-archive core, together with its three components, the transit network, metro-wide job growth and job relocation, which sum to the observed line in every year. Measured against the 2014 annual average, and in the

Figure 4: Network, job-growth, and job-relocation components of the change in job access by transit, 2014–2025

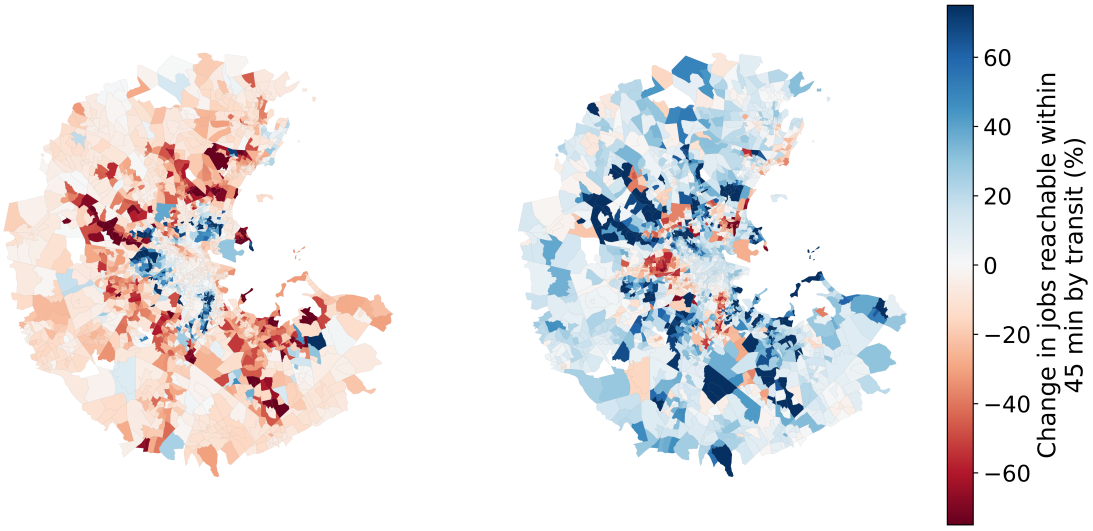


Note: Decomposition of the change in job access by transit into the transit network (jobs held fixed), metro-wide job growth, and job relocation components, corrected for archive entry, within block group, annual, population-weighted, the clean-archive core (19 metros outside New York), indexed to the 2014 annual average; vertical axis in log-points, the units in which the three components sum to the observed change. Shading is the 95% block-group-bootstrap band.

log-point units in which the components add, metro job growth alone would have added +15.9 log-points to the typical neighborhood's job access by transit as of 2025. The transit network's own contribution holds within a point of zero through 2019 and then falls sharply after the pandemic, ending at −6.4 log-points. Job relocation contributes +5.7 log-points. Employment moved toward transit-served locations, offsetting most of the network's negative contribution. The three sum exactly to the observed +15.2 log-points, a rise of about 16.4%, on this sample. The 52-metro trajectory of Section 5.1 ends the decade at +14.1 log-points (+15.1%). Table A.7 reconciles the two in three steps, the largest of which is the sample itself, because the clean-archive metros grew faster. That step changes the observed rise but not the split among the components. Metro job growth and job relocation survive clustering at the metro level, and so does the net decade change (Table A.6). They are common processes across the 52 metros. The transit network component, by contrast, has a metro-clustered interval that spans zero; metros' network paths diverge (Section 5.4). In the decomposition, the rise in jobs reachable by transit since 2014 is accounted for by employment growth, not by transit service. Jobs are observed through 2023, so 2024–25 movement is network-only by construction; a decomposition ending at 2023 reads nearly identically (observed +14.9, job growth +15.9, relocation +5.7 and network −6.6, each rounded).

5.3 The pandemic

Figure 5: Block-group change in job access by transit, Boston, 2019Q4–2025Q4



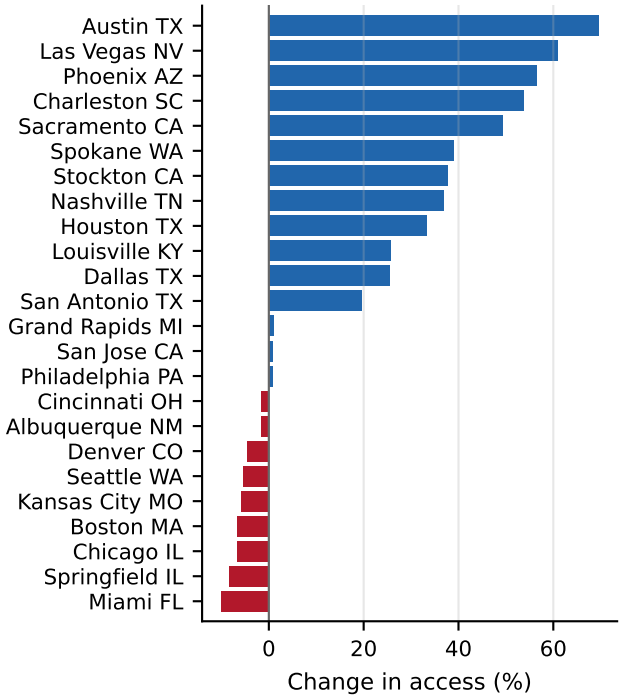
Note: Percent change in jobs reachable within 45 minutes by transit, per block group, Boston: the collapse (left panel, 2019Q4 to 2020Q3) and the recovery (right panel, 2020Q3 to 2025Q4), shared diverging scale, blue for gains and red for losses (central 30 km; base $\geq$ 100 jobs).

COVID-19 is the episode the panel resolves best. The 2020 collapse and recovery are visible in the trajectory (Fig. 3) and in the decomposition (Fig. 4). From 2019 to 2020 the network component and the job-growth component each fell about four log-points. Job growth was back above its 2019 level by 2022, while the network component did not return to its 2019 level. The network component's fall is the job-access counterpart of the 2020 service cuts (Kar et al., 2022). We compare the component's incomplete recovery with agency-reported service data (the National Transit Database), although their path through the collapse differs (Fig. A.3). In the clean-archive core, vehicle revenue miles (the miles transit vehicles travel in passenger service, at all hours) fell about 14% from 2019 to 2020, against about 4% for the network component. Service miles then held near their 2020 level through 2023, while the network component kept falling until 2022. All-day service miles returned faster than peak-hour job reach. The comparison is not circular. A divergence from the agency data is flagged only when the archived feeds confirm the archive as its cause (Table A.2). In nine of the 19 core metros the two series differ by 12 log-points or more for two or more consecutive years without that confirmation. The core is therefore not selected for agreement, and selection toward agreement could not produce the pre-2020 disagreement the exhibit shows, when service miles rose about 11% while the network component stayed near zero. The panel adds the collapse and the recovery of job access at the neighborhood level, quarter by quarter, within the longer decade.

Figure 5 shows the collapse and the recovery as two change maps for Boston, chosen because its archived schedules are dense and unflagged from 2019Q4 to 2025Q4. Both maps combine schedule changes with changes in the job counts, from the 2019 to the 2020 LODES year in the collapse and from 2020 to 2023 in the recovery. The collapse (left) is broad but deepest in the middle-ring neighborhoods. The recovery (right) reverses it, except in the neighborhoods shown in red, where access in 2025Q4 was below its 2020Q3 level. The mapped area's median reachable-job count fell by about a fifth (−22%) in three quarters; an annual measure would average over the entire movement.

5.4 Across metros: winners and losers

Figure 6: Decade change in job access by transit across metropolitan areas



Note: The typical neighborhood's change in job access by transit, 2014–15 to 2024–25, by metro (corrected, within block group, population-weighted; top and bottom 12 of the 45 of 52 metros with both endpoint windows observed; the seven unranked metros, without a clean window at one or both endpoints, are Bremerton, Los Angeles, Oklahoma City, Providence, Redding, Salinas, and Washington). Job counts for Grand Rapids are held at their 2021 values after 2021 (Section 3.1).

The trajectory averages over wide variation across metros, which the panel can rank because its method is held fixed (Fig. 6). The figure ranks the typical neighborhood's decade change in each metro, from the mean of 2014–15 to the mean of 2024–25, and shows the twelve largest gains and the twelve smallest changes, nine of them declines, among the

45 metros observed at both ends. The largest gains accrued to fast-growing Sun Belt and western metros (Austin, Las Vegas, Phoenix, Charleston, and Sacramento lead), while the five largest declines were in Miami, Springfield (Illinois), Chicago, Boston, and Kansas City.

The correction matters here. On the ranking's window-mean construction, Miami's uncorrected series shows a +18% gain that is entirely operators entering the archive after 2015. Once these entries are removed, the typical Miami neighborhood lost about 10%, the largest measurable decline (Fig. A.1 traces the same correction quarter by quarter against the 2014 mean). Chicago, whose post-2020 schedules were recovered (Section 3.3), declined about 7%.

5.5 Within cities: who has job access by transit?

Levels and growth give opposite answers, and we take them in turn. In levels (as of 2023), job access by transit is highest in the neighborhoods that rely on it. For each characteristic, Table 2 compares the neighborhoods at the two ends of its range, Group 1 and Group 4, so a ratio above one means Group 1 reaches more jobs. Population-weighted on the panel, the poorest within-metro income quartile reaches about 1.9× the access of the richest; neighborhoods where more than 30% of households are car-free reach about 14× the access of car-owning ones; high-poverty neighborhoods reach about 2.4× low-poverty ones; and central neighborhoods reach roughly 50× the urban fringe. Outside New York, neighborhoods more than 60% Black reach more than those less than 10% Black (about 2.1×), while neighborhoods more than 60% Hispanic reach about the same as those less than 10% Hispanic (1.0×). Each ratio's confidence interval excludes one, except the Hispanic contrast, whose interval includes one. A natural objection is that these gradients merely restate urban form, because poor, car-free, and minority households live centrally, where transit is. Controlling for distance to the urban core separates the two. We compare neighborhoods within four bands of distance from the nearest employment center (0–5, 5–15, 15–30, and more than 30 km). The car-reliance gradient survives within every band. Car-free neighborhoods reach more jobs by transit than their car-owning neighbors at the same distance from the center. The income and poverty gradients also survive within every band, strongest in the 5–30 km suburban ring and attenuated but still pro-poor in the dense core. The racial gradient, by contrast, is near one within distance bands. The advantage of neighborhoods more than 60% Black in the pooled comparison reflects where those neighborhoods sit (closer to the core) rather than race per se.

The transit/car gap is large. Outside New York in 2023, the population-weighted average neighborhood reaches roughly thirty-eight-fold fewer jobs by transit than by car within 45

Table 2: Job access by transit across neighborhood groups

Group contrast (G1 / G4)	G1	G4	Ratio	95% CI
Income: poorest / richest	53.4	28.0	1.90	[1.77, 2.05]
Car-free: >30% / <10%	240.1	16.9	14.21	[13.54, 14.92]
Poverty: >60% / <20%	72.2	29.7	2.43	[2.28, 2.59]
Distance: core / fringe	124.0	2.4	50.75	[48.70, 52.89]
%Black: >60% / <10%	71.5	33.6	2.13	[1.99, 2.28]
%Hispanic: >60% / <10%	37.0	37.9	0.98	[0.92, 1.04]

Note: 45-minute budget, 2023, 52 metros outside New York, in thousands of jobs. Group 1 and Group 4 access (population-weighted means of block-group 2023 values) and their ratio with a delta-method 95% confidence interval, all jobs (C000); each ratio is Group 1 over Group 4, so a ratio above one means Group 1 reaches more jobs (ratios computed before rounding). The distance groups are the innermost and outermost distance-to-core quartiles, formed within each metro.

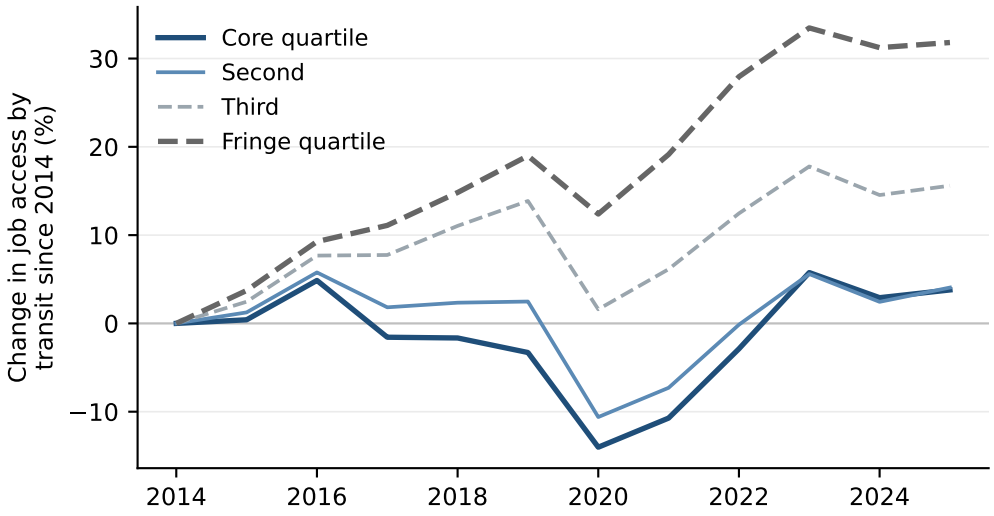
minutes. In the median neighborhood, a car reaches about 174 times as many jobs as transit. The two summaries differ because access is severely right-skewed, and the ratio of population-weighted means is pulled toward the transit-rich core, where transit does comparatively well. Table A.4 reports the gap at each time budget, from 86× at 15 minutes to about twentyfold within an hour.

5.6 Within cities: who gained access?

Growth ran in the opposite direction from levels, and on the corrected index the difference is large. Figure 7 follows the four distance-to-core quartiles. The urban fringe gained about 32% over the decade while the core gained little (+4%). The outer two quartiles had already gained 14 to 19% by 2019, and only the inner two fell below their 2014 level in 2020.

Figure 8 shows the decade change for income, poverty, car-ownership and racial groups. Filled markers are the corrected estimates and open markers the uncorrected values, so the distance between them is the archive adoption the correction removes. In the income, poverty and car-ownership panels, the better-off group gained the most. The richest income quartile gained about 31% while the poorest gained essentially nothing (−0.3%); the most car-owning neighborhoods gained about 19% while the most car-free gained essentially nothing; low-poverty neighborhoods gained about 22% while high-poverty neighborhoods gained barely a point. The largest difference is by race. A neighborhood is majority-White, majority-Black or majority-Hispanic when more than half of its residents are, respectively, non-Hispanic White, non-Hispanic Black or Hispanic. Majority-White neighborhoods gained about 16% (majority-Hispanic about 11%), while majority-Black neighborhoods' job access by transit declined, by about 9% within 45 minutes and by roughly 13% within 60 minutes. The

Figure 7: Change in job access by transit across distance-to-core quartiles, 2014–2025

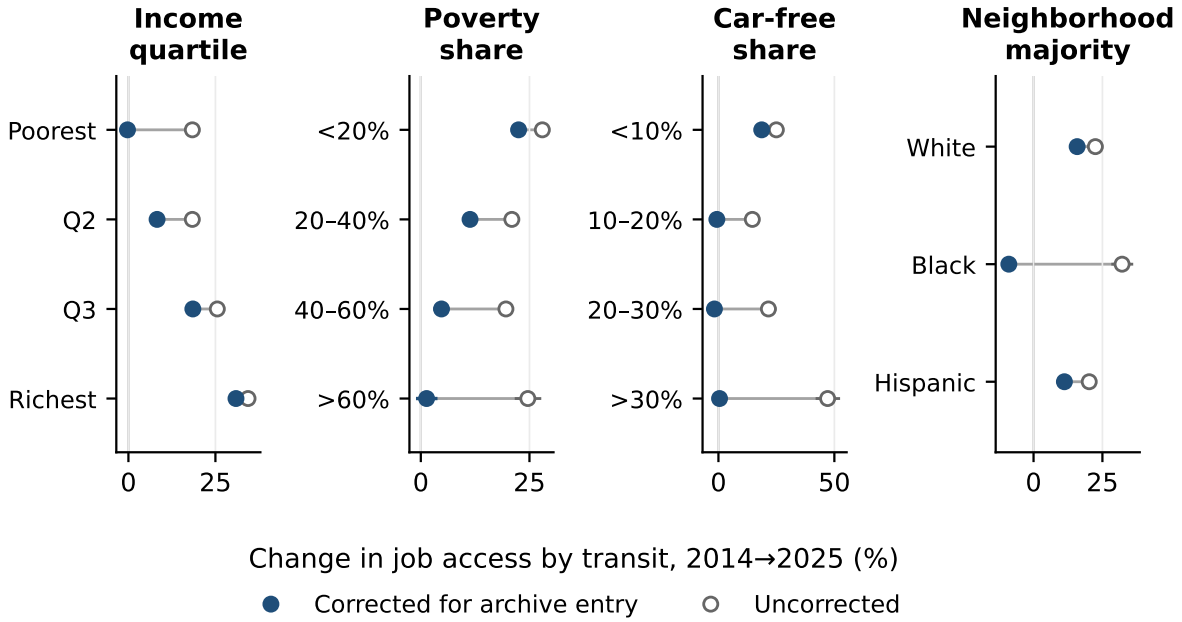


Note: Within-block-group change in job access by transit relative to 2014, corrected for archive entry, by distance-to-core quartile (quartiles formed within each metro), annual, population-weighted, 52 metros outside New York, flagged cells excluded.

decline is concentrated in the handful of metros where most residents of majority-Black neighborhoods live, so it is a fact about those metros rather than an estimate of a uniform national process (at both budgets its metro-clustered interval spans zero; Table A.6). The gains in richer, whiter, peripheral neighborhoods, by contrast, hold under metro clustering. The within-city gaps opened before the pandemic. Compared year by year within the same metro and the same distance band as in Section 5.5, the poorest–richest contrast had already reached roughly −20 log-points by 2019. The gradient is not an artifact of the 45-minute budget. It holds at 30 and 60 minutes, steeper at 30 and shallower at 60 (Table A.8). The majority-Black decline grows with the budget. Within 30 minutes the change is +0.5%, and longer-reach access, which covers more of the metro's jobs, fell the most. The decline appears only on the corrected series, and on the uncorrected series these neighborhoods gained. The operators that entered the archive after 2014 disproportionately serve poorer and majority-Black neighborhoods, so the two readings differ most for them.

The mechanism is the one the decomposition identifies (Section 5.2). The movement is chiefly on the jobs side, with reachable-job counts growing where jobs grew and falling where they declined (Table 3). The decomposition's relocation component shows where jobs moved. It is slightly negative in the two inner distance quartiles (about −1 log-point) and strongly positive at the fringe (+16), concentrated in fast-growing metros. Employment moving outward leaves the set of jobs reachable from the core and enters the sets reachable from mid-ring and peripheral neighborhoods. Summed over all neighborhoods, this outward

Figure 8: Decade change in job access by transit across neighborhood groups



Note: Change in job access by transit from 2014 to 2025 by neighborhood group (within-block-group fixed-effects estimates for 2025 against the 2014 average), population-weighted, 52 metros outside New York, flagged cells excluded. Filled markers: corrected for archive entry; open markers: uncorrected; whiskers: 95% block-group-clustered intervals. The distance between the two is the archive-entry steps the correction removes.

movement counts as relocation toward transit-served places, which is why relocation adds to access overall while the core gained little. In percentage terms low-access peripheral neighborhoods converged toward high-access central ones; in absolute terms every change is small against the level differences (the fringe's +32% is under a thousand reachable jobs; the core's few-percent gain, several thousand). We report both readings and attach no welfare claim to either.

The growth gradient is not a composition of growing metros, nor of urban form alone. Table 3 compares each neighborhood only with others in the same metro and the same five-kilometer band of distance from the metro's block group with the most jobs. Each row is a contrast between two groups of block groups, and the columns split the observed difference in decade change into its network and jobs components, in log-points. The income gradient survives essentially intact. The poorest quartile's access grew 20.7 log-points less than the richest quartile's over the decade (95% CI $[-22.1, -19.2]$; metro-clustered $[-26.8, -14.5]$, robust at either level). The poverty gradient is similar. The racial gap attenuates by about two-thirds, location explaining most of it, but a majority-Black shortfall of about 11 log-points remains ($[-13.0, -8.3]$; under metro clustering it spans zero, Table A.6). Decomposing

Table 3: Decade change in job access by transit within cities, by neighborhood contrast

Contrast (within metro × distance band)	Observed	Network	Jobs
Poorest vs. richest income quartile	−20.7	−2.5	−18.8
	[−22.1, −19.2]	[−3.5, −1.5]	[−19.9, −17.7]
High- vs. low-poverty (>60% vs. <20%)	−17.9	−3.8	−15.4
	[−19.9, −15.9]	[−5.4, −2.2]	[−16.8, −14.1]
Majority-Black vs. majority-White	−10.7	−7.0	−5.1
	[−13.0, −8.3]	[−8.9, −5.0]	[−6.7, −3.4]
Car-free vs. car-owning (>30% vs. <10%)	−5.7	−0.5	−5.8
	[−7.6, −3.8]	[−2.4, +1.3]	[−6.7, −5.0]

Note: In log-points. Each cell is the coefficient on the listed contrast from a population-weighted regression of each block group's decade change on group indicators, with metro × distance-band cells absorbed. The decade change is the 2024–25 mean of log access minus the 2014–15 mean, corrected for archive entry, and the bands are 5 km wide, by distance from the metro's block group with the most jobs. Brackets give robust 95% confidence intervals. The sample is 70,057 block groups in the 45 metros observed in both 2014–15 and 2024–25. Network and Jobs are the decomposition's components (Section 4.3) on this sample, measured against the 2014 average, with the correction applied to Network as to Observed.

each contrast into its network and jobs components, with the archive-entry correction applied equally to the observed change and the network part so the columns approximately add, shows which component carries each gap. For income, poverty, and car-reliance the gradient is on the jobs side, with jobs-side losses of −19, −15, and −6 log-points against network components within a few points of zero. The racial contrast splits. About half is on the jobs side (−5.1 log-points; the jobs part holds the network fixed, so archive entries do not affect it), while the sign of the network part cannot be determined (it is −7 log-points if every archive entry is treated as adoption and +7 if every entry is genuine new service), so we do not interpret it. Within the same city and the same distance from the core, employment relocated over the decade toward places reachable from richer neighborhoods. For the poorest-versus-richest contrast, transit service change played essentially no role. Because neighborhoods are classified at a fixed 2018–2022 vintage, this reads as "the places that are affluent today gained access"; whether jobs moved toward richer neighborhoods or richer households moved toward access is a sorting question this comparison does not resolve. One version of the concern we can address is end-of-period sorting. Classifying neighborhoods instead by their income at the start of the period, from the 2009–13 ACS crosswalked to 2020 block groups (Schroeder et al., 2026), leaves the gradient essentially intact (Table A.8). The neighborhoods that were already better-off at the start of the period are the ones whose access grew; the pattern is not an artifact of who is rich today.

5.7 The transit-reliant margin

This section studies car-free households in low-income neighborhoods. Table 4 reports the panel for this group, using constrained transit-reliant strata. Its rows are all block groups and three constrained strata, defined by thresholds on the car-free and poverty shares, and its columns give each stratum's size, the 2023 access of its typical car-free household, the corrected change over the decade, and the share of its car-free households in its three largest metros. All statistics weight block groups by their car-free households, so each row describes the typical car-free household in that stratum. In levels, the sorting documented cross-sectionally by the Modal Access Gap literature (Maharjan et al., 2024) is visible on the constrained margin too. In 2023 the typical car-free household lived where transit reached 139,000 jobs within 45 minutes, roughly 3.7 times the population-weighted average person's 38,000 (Table A.4). Even the poorest, most car-free strata reached 138–191,000, more as the thresholds rise. Two caveats apply. Membership is fixed at the 2018–2022 ACS vintage, so the strata are today's reliant neighborhoods traced backward. Classifying them on their 2009–13 attributes instead gives the same answer (Table A.8), so the small gain for car-free households is not an artifact of end-of-period sorting. The strata are also concentrated in a few dense metros (Chicago, Los Angeles, and Miami together hold about 31% of the constrained car-free weight), and New York, the most extreme case, is analyzed separately.

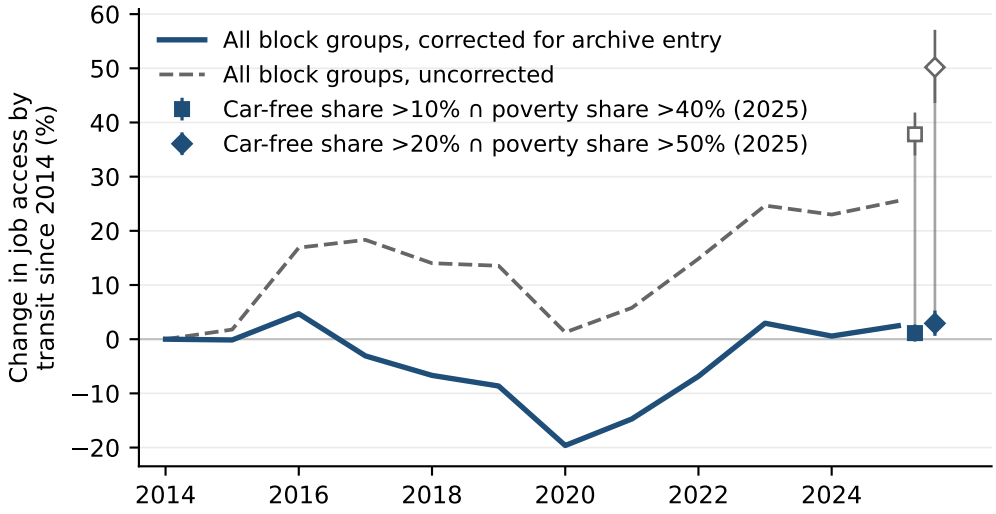
Table 4: Job access by transit at the transit-reliant margin

Stratum	Block groups	Car-free households covered (%)	Access 2023 (000s)	Change 2014–25 (%)	Top-3 metro share (%)
All block groups	85,694	100	139	+2.5	29
Constrained: car-free share ∩ poverty share					
>10% ∩ >40%	10,400	38	138	+1.1	31
>20% ∩ >50%	4,386	22	159	+2.9	34
>30% ∩ >50%	2,585	16	191	+2.8	38

Note: 52 metros outside New York, 2014–2025, flagged cells excluded. Block groups and car-free coverage count each block group once over 2014–2025. Access = jobs reachable within 45 minutes by transit in 2023 (thousands, all jobs), weighted by car-free households (ACS 2018–2022: car-free from B25044, poverty share from C17002; membership fixed at that vintage). Change 2014–25 = the stratum's archive-entry-corrected within-block-group fixed-effects estimate for 2025 (Fig. 9). Top-3 metro share = the share of the stratum's car-free households living in its three largest metros. The thresholds apply to neighborhood shares, not to households (Section 3.1).

Over time, the decade's rise is small for this group. Figure 9 plots the change for the average car-free household, corrected (solid) and uncorrected (dashed), and at right the decade change for the two broadest constrained strata, with filled markers corrected and open markers uncorrected. On the index, weighted by car-free households, access for the

Figure 9: Change in job access by transit, weighted by car-free households, 2014–2025



Note: Change in job access by transit relative to 2014, annual, weighted by car-free households, all block groups, 52 metros outside New York, flagged cells excluded: the corrected series (solid) and the uncorrected series (dashed); the gap is the archive-entry steps the correction removes. At right, the 2025 decade change for the two broadest constrained strata (car-free share >10% ∩ poverty share >40%, squares; car-free share >20% ∩ poverty share >50%, diamonds): filled markers corrected, open markers uncorrected, whiskers 95% block-group-clustered.

average car-free household rose just +2.5% over the decade (CI [+1.6, +3.3]), a small fraction of the +15% the average resident gained, and the constrained strata gained no more, ending between +1 and +3% above their 2014 level (block-group intervals that straddle or barely clear zero). The uncorrected series ends far higher (about +26%); the gap between the two is precisely the operators entering the archive in these neighborhoods, the adoption the correction exists to remove (Section 4.2). The small gain is the corrected reading, and the distance-controlled comparison supports it independently. These strata start high in levels, being central, but they gained little over the decade. The more informative comparison is relative and survives the urban-form control. Comparing block groups within the same metro and the same five-kilometer distance band (the framework of Table 3), constrained neighborhoods' decade growth was 8 to 13 log-points lower than their same-distance neighbors' (metro-clustered intervals exclude zero). The shortfall is chiefly jobs-side (the jobs component alone is about 8 log-points, while the network component is small and spans zero across archive-entry treatments). In the decomposition, the shortfall is accounted for by employment relocating away from these neighborhoods, not by service withdrawal or by distance to the core. This is the within-city result of Section 5.6, measured for car-free households.

6 New York: the exception

New York is not a large version of the other metros. We report it on its own (Section 3.2); Zhang et al. (2026) give a dedicated account of how New York's job access by transit and its racial distribution evolved, and here we compare New York with the other metros. Four facts set it apart.

Scale. The average New York resident reaches more than ten times the jobs by transit of the average resident elsewhere, about 410,000 within 45 minutes in 2023 against 38,000 outside New York (Table A.4). Its transit/car gap is correspondingly small. A car reaches about 12 times as many jobs as transit within 45 minutes there, against 38 times elsewhere, because transit in New York is competitive with the car.

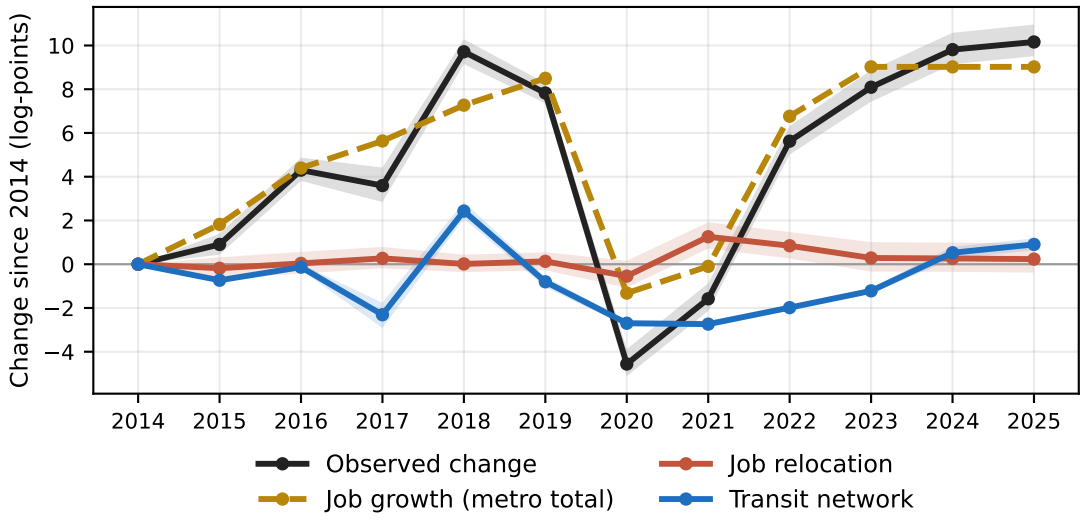
Trajectory and the pandemic. Measured within block group, New York's job access by transit rose about +11% over the decade against +25% outside it. Both figures are on the uncorrected index. With a single metro, every block group changes its set of archived operators in the same quarters, so the archive-entry correction cannot separate an entry step from the trend, and a corrected New York index is near zero by construction. New York's access dipped below the 2014 baseline in 2020 (about −5%), as it did outside New York.

The decomposition. New York's network component ended the decade near its 2014 level, while the clean-archive core's fell. Figure 10 plots New York's decomposition in the same form as Fig. 4. Like its trajectory, it carries no archive-entry correction. Holding jobs fixed, New York's network adds about +1 log-point over the decade. On the same uncorrected basis the clean-archive core's network component fell about 5 log-points (6 with the correction). New York's rise is almost purely job growth (about +9 log-points; net job relocation is nil at +0.2).

The gradients. Outside New York, job access by transit is higher in poorer neighborhoods and in neighborhoods more than 60% Black (Table 2). In New York the income gradient disappears and the racial gradient reverses (Table 5). Figure 11 maps job access by transit across the region with its majority-Black block groups outlined. Most of them are in Brooklyn, southeast Queens, Newark and the Bronx, away from the access peaks rather than on them.

New York's poorest and richest income quartiles reach about the same access (1.08×, with a confidence interval that includes one). Neighborhoods more than 60% Black reach about a third fewer jobs than those less than 10% Black (0.64×), the opposite of the roughly

Figure 10: Network, job-growth, and job-relocation components of access change: New York



Note: The decomposition of Fig. 4 for New York alone, without the archive-entry correction: transit network (jobs held fixed), job growth, and job relocation components, within block group, annual, population-weighted, log-points, indexed to the 2014 annual average. Shading is the 95% block-group-bootstrap band.

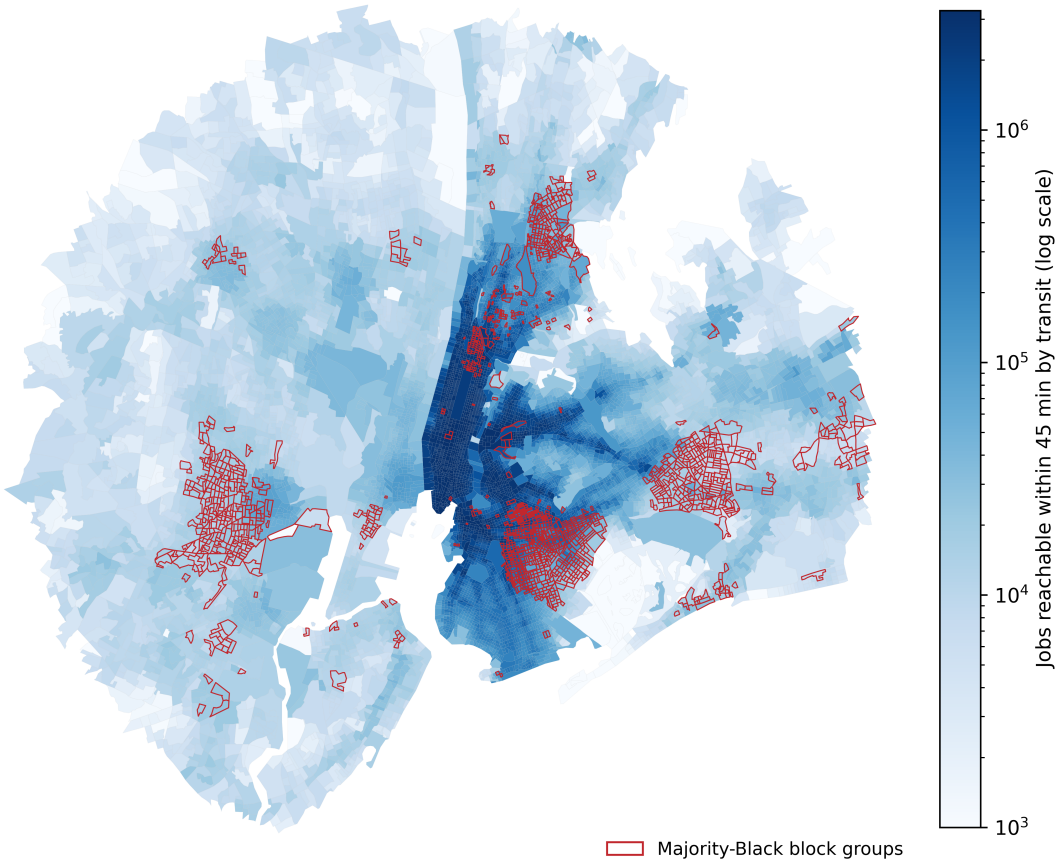
Table 5: New York: job access by transit across neighborhood groups

Group contrast (G1 / G4)	G1	G4	Ratio	95% CI
Income: poorest / richest	486.6	448.7	1.08	[0.98, 1.20]
Car-free: >30% / <10%	1,026.9	10.6	96.86	[89.38, 104.97]
Poverty: >60% / <20%	603.1	328.7	1.83	[1.64, 2.05]
Distance: core / fringe	1,330.8	4.7	284.79	[270.97, 299.32]
%Black: >60% / <10%	260.8	406.4	0.64	[0.56, 0.73]
%Hispanic: >60% / <10%	474.8	368.6	1.29	[1.16, 1.43]

Note: 45-minute budget, 2023, New York, in thousands of jobs. Group 1 and Group 4 access (population-weighted means of block-group 2023 values) and their ratio with a delta-method 95% confidence interval, all jobs (C000). Groups and their order are as in Table 2.

2.1× advantage outside New York. The car-reliance and distance gradients, by contrast, are even steeper than elsewhere (about 97× and 285×).

Figure 11: Job access by transit per block group, New York, 2023Q4



Note: Jobs reachable within 45 minutes by transit, per block group, 2023Q4 (log scale, central 35 km), with the 1,513 majority-Black block groups in the mapped area outlined.

7 Discussion and Conclusion

Between 2014 and 2025, the typical neighborhood in the 52 metros outside New York gained about 15% in job access by transit. Access fell below its 2014 level during the 2020 service cuts and ended the period above its pre-pandemic peak. The decomposition shows where the gain came from. In the clean-archive metros, growth in the number of jobs alone would have produced more than the entire rise. The network component (the change in access from the schedule alone, with jobs held fixed) contributed close to nothing before 2020 and subtracted after it. The movement of jobs toward places that transit serves offset most of that loss. The size of the network's post-2020 decline is not precisely measured, because metros differ widely and the agency-reported all-day service we compare it with recovered faster than peak job reach (Table A.6, Fig. A.3). Its near-zero contribution before 2020 and its 2020 collapse are clear. The gains were not shared evenly. Access is highest in poor,

central, car-free neighborhoods, but the growth went mainly to higher-income, car-owning neighborhoods at the edges of metros, where jobs grew. Weighted by car-free households, access rose 2.5%, compared with 15% for the average resident, and on the corrected series majority-Black neighborhoods, concentrated in a few metros, ended the decade below their 2014 access. In New York, analyzed separately, the network component ended the decade near its 2014 level (Section 6).

These results describe how transit networks and the geography of employment interacted over the decade. Access rose where new jobs located within reach of existing service, mostly in the middle and outer rings of fast-growing metros, and changed little in the central neighborhoods that already had the most access. The distributional results follow from that geography. The neighborhoods that rely most on transit are central, where job growth added little to an already large set of reachable jobs, and within the same city and distance from the core, jobs moved toward places reachable from richer neighborhoods. A rise in average access can therefore coexist with little or no gain for the residents who depend on transit.

The results have three implications for practice. First, agencies and funders that track recovery through service hours or vehicle revenue miles can overstate the recovery of access to work. In the clean-archive metros, all-day vehicle revenue miles ended the decade about 7% above their 2014 level, while the network's contribution to morning-peak job access by transit ended about six log-points below zero. Job access by transit can be computed from an agency's own published schedules, so it can be reported beside service hours. Second, the location of employment is part of policy on job access by transit. For regional plans that set access to jobs as a goal (Section 2), the goal depends on where employment grows as well as on transit service, and land-use decisions are part of how it is met. Third, equity analysis should measure access and weight it by the people it concerns. Federally required service equity analyses examine changes in service, but the within-city gaps by income, poverty and car ownership that opened over this decade came almost entirely from the movement of jobs (Table 3). Averages over all residents also overstate what transit-dependent households experienced (Section 5.7).

The results have several limitations. GTFS records scheduled rather than delivered service, so the panel measures timetabled access rather than the reliability a rider experiences. The panel covers only the morning peak. A constant peak-hour measure tracks relative accessibility across neighborhoods closely (Boisjoly and El-Geneidy, 2016), but off-peak and weekend access matter most for non-standard work schedules. Cumulative access counts reachable jobs. It does not account for competition for those jobs, fares, crowding, safety or

transfers, so it measures potential rather than effective access, and the omission of competition matters most for the low-wage workers most dependent on transit (Shen, 1998). LODES records a job at its employer's establishment, so remote work lowers the relevance of physical access for telework-able jobs after 2020 (Dingel and Neiman, 2020). Reweighting the jobs side to work that cannot be done remotely leaves the relocation finding intact. Finally, the archive covers larger, transit-intensive metros best, and where operators entered the archive late, mainly in majority-Black and car-free neighborhoods, the corrected and uncorrected readings differ widely (Figs. 8 and 9).

The analysis is descriptive. It shows where the jobs a rider can reach came from, not whether reaching them changed anyone's employment. The panel provides a longitudinally comparable measure of job access by transit, quarterly and by neighborhood, that causal studies of access and employment can use. Such trend work on archived schedules needs auditing against agency-reported service, as Section 3.3 shows.

Over the past decade, the number of jobs residents of these metros could reach by transit rose mainly through employment growth and relocation, while the transit network added little and then subtracted. The residents who depend on transit gained the least. Whether transit connects households without a car to work depends on where jobs locate as well as on where and how often transit runs. A policy that tracked only transit service would have missed most of what changed.

Data availability

The panel is constructed entirely from publicly available sources: archived GTFS feeds from the Transitland archive, the Mobility Database and the published archive of the Southeastern Pennsylvania Transportation Authority; LODES employment counts; OpenStreetMap street networks; and American Community Survey tables. The checks also use the National Transit Database and the published *Access Across America* values. Sections 3 and 4 describe the sources and their processing.

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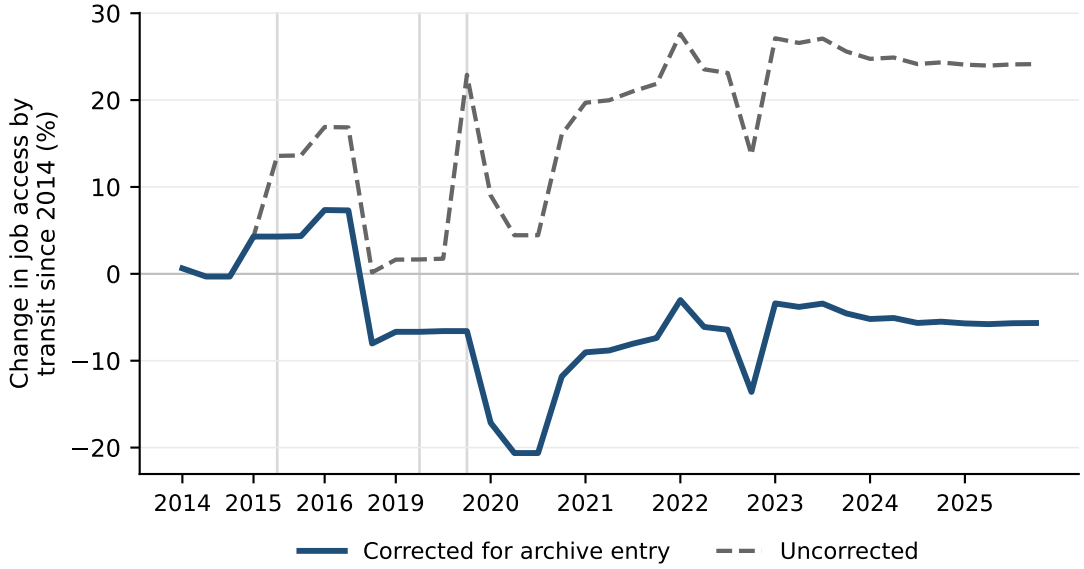
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A Supporting exhibits

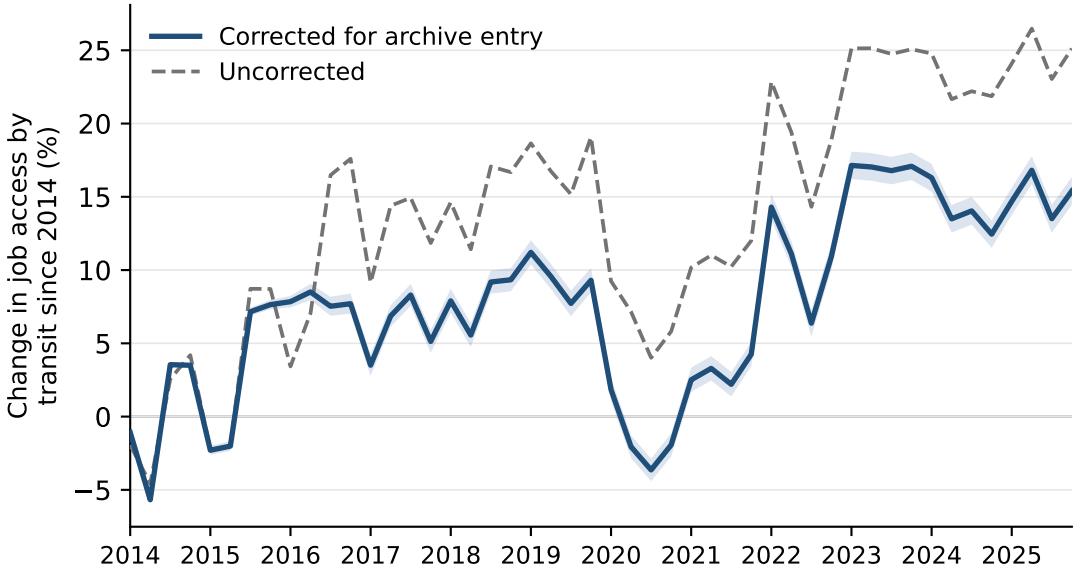
Figures

Figure A.1: The archive-entry correction in one metro: Miami



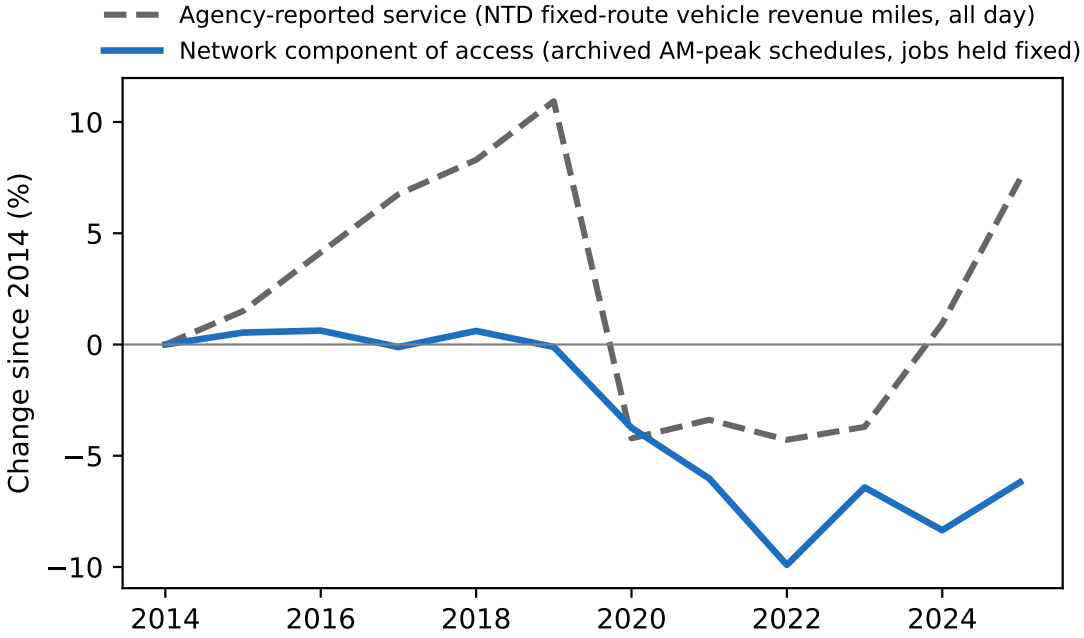
Note: Population-weighted mean job access by transit in Miami, uncorrected (dashed) and with the archive-entry correction applied per block group (solid), percent change against the 2014 annual mean, over the 3,942 block groups present in every covered quarter (the series starts in 2014Q2, and 2015Q2 and 2016Q3–2018Q3 are uncovered; the axis spaces covered quarters evenly). Vertical lines mark changes in Miami's archived operator set; the separations between the two series at those marks are the archive-entry steps the correction removes.

Figure A.2: Change in job access by transit relative to 2014, corrected and uncorrected, 2014–2025



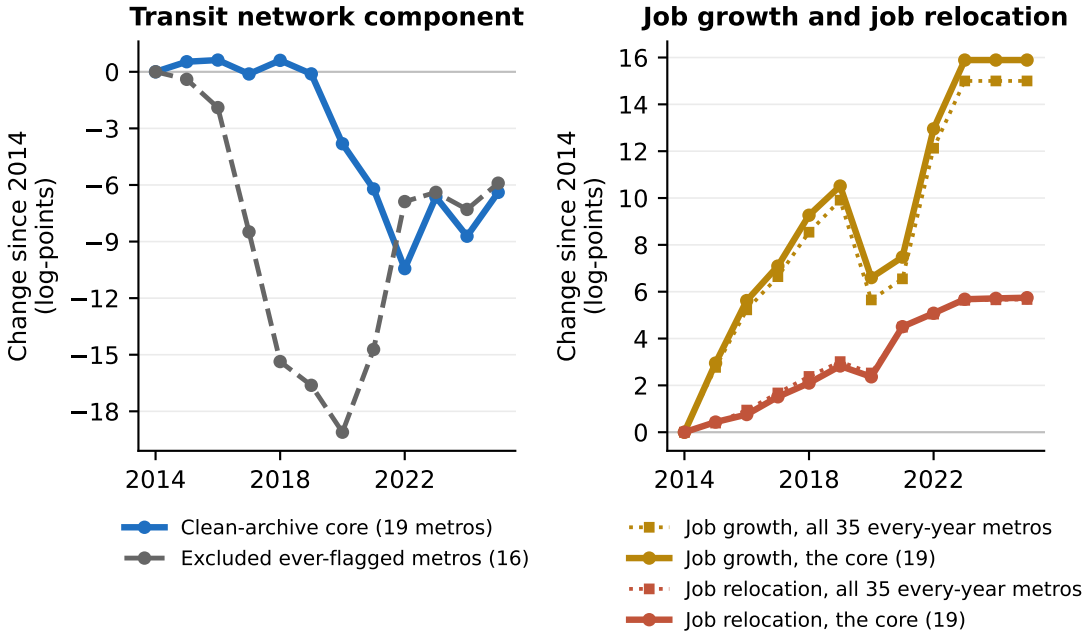
Note: Within-block-group fixed-effects change in jobs reachable within 45 minutes by transit, quarterly, relative to the 2014 average, population-weighted, 52 metros outside New York. The solid line is the archive-entry-corrected index of Fig. 3; the dashed line is the uncorrected series. Shading is the 95% block-group-clustered band on the corrected index.

Figure A.3: Transit-network component of access and NTD vehicle revenue miles, 2014–2025



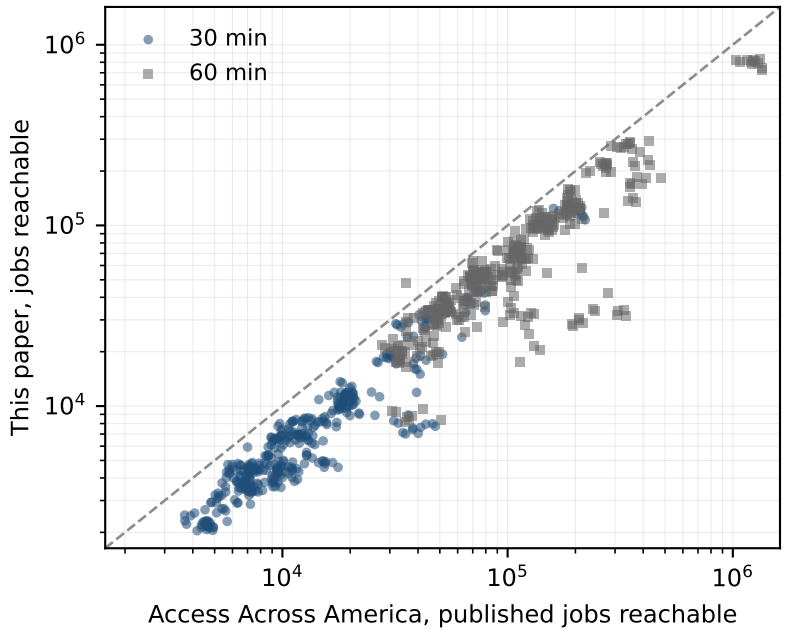
Note: Change since 2014 in agency-reported fixed-route vehicle revenue miles (VRM, from the National Transit Database, the standard measure of service supplied, at all hours; dashed) and in our transit-network component (archived Wednesday-AM-peak schedules, jobs held fixed; solid), pooled over the clean-archive core (19 metros), all with complete 2014–2025 NTD coverage. Both series are in percent, the network component converted from log-points.

Figure A.4: Decomposition component paths by metro archive status



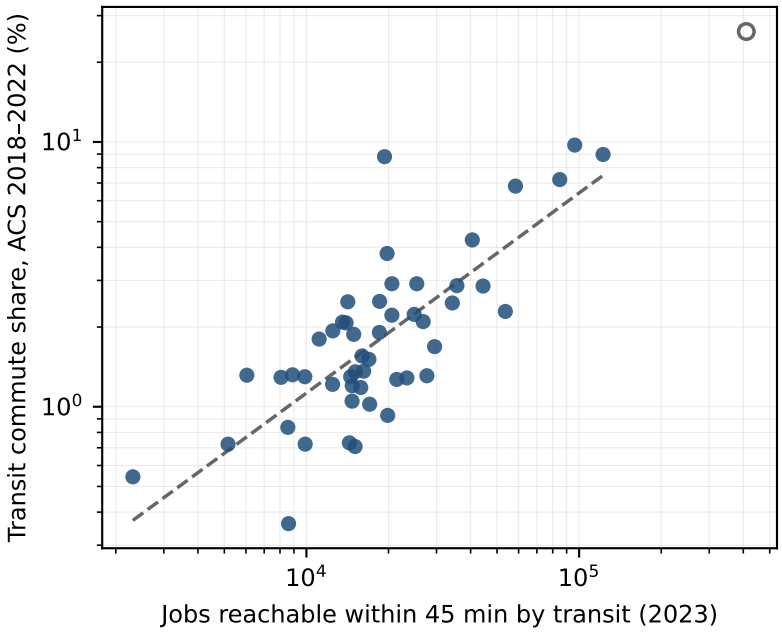
Note: The left panel shows the transit-network component on the clean-archive core (19 metros, solid) and on the 16 every-year metros excluded for flagged quarters (dashed); corrected, within block group, population-weighted, log-points vs the 2014 annual average. The right panel shows the job-growth and job-relocation components on all 35 every-year metros (dotted) and on the core (solid), same construction. A metro with any flagged quarter is excluded whole (Section 4.3). The core's 19 metros are Albuquerque, Austin, Champaign–Urbana, Charleston, Chicago, Cincinnati, Denver, Kansas City, Madison, Oxnard–Thousand Oaks, Phoenix, Portland, Rochester, Sacramento, San Antonio, Santa Rosa, Seattle, Spokane, and Stockton, which hold 33% of the panel's population. Grand Rapids (2014), Nashville (2025), Redding (2024–25) and Salinas (2014–15) have no flagged quarter but no quarter at all in the years shown, and are not among the 35 every-year metros.

Figure A.5: Access levels against published *Access Across America* values



Note: Our population-weighted metro-year access against the worker-weighted values published in the *Access Across America: Transit* reports (Owen et al., 2016, 2025), at the 30- and 60-minute thresholds, log scales; the dashed line is equality. The 390 overlapping metro-years cover 43 metros, 2014–2024. The published values cover the urban area and ours the larger metropolitan area, and departure-time protocols and employment vintages also differ; flagged cells are excluded on our side (Section 3.3).

Figure A.6: Metro job access by transit against the ACS transit commute share



Note: Metro job access by transit (2023, flagged cells excluded) against the ACS 2018–2022 transit commute share (table B08301), log scales. The 51 metros are the panel and New York (hollow marker), less Bremerton and Los Angeles, which have no unflagged 2023 quarter. The dashed line is the least-squares fit in logs outside New York.

Tables

Table A.1: The 52 panel metros and quarters present (of 48, 2014–2025)

Metro	Qtrs	Clean	Metro	Qtrs	Clean
Albany, NY	48	47	Milwaukee, WI	48	46
Albuquerque, NM	48	48	Nashville, TN	39	39
Anchorage, AK	46	45	Oklahoma City, OK	38	35
Atlanta, GA	46	38	Oxnard, CA	44	44
Austin, TX	46	46	Philadelphia, PA-NJ	48	45
Boston, MA-NH	48	46	Phoenix, AZ	48	48
Bremerton, WA	48	28	Portland, OR-WA	47	47
Champaign, IL	47	47	Providence, RI-MA	39	33
Charleston, SC	48	48	Redding, CA	39	39
Chicago, IL-IN	44	44	Reno, NV	48	47
Cincinnati, OH-KY-IN	48	48	Rochester, NY	47	47
Cleveland, OH	46	44	Sacramento, CA	47	47
Columbus, OH	42	38	Salem, OR	46	35
Dallas, TX	48	42	Salinas, CA	37	37
Denver, CO	48	48	Salt Lake, UT	44	43
Duluth, MN-WI	48	46	San Antonio, TX	45	45
Eugene, OR	42	41	San Diego, CA	46	44
Grand Rapids, MI	44	44	San Jose, CA	46	44
Houston, TX	48	47	Santa Cruz, CA	40	38
Indianapolis, IN	43	37	Santa Rosa, CA	48	48
Kansas City, MO-KS	45	45	Seattle, WA	45	45
Las Vegas, NV	43	40	Spokane, WA	48	48
Los Angeles, CA	39	27	Springfield, IL	42	41
Louisville, KY-IN	37	35	Springfield, MA	46	32
Madison, WI	48	48	Stockton, CA	46	46
Miami, FL	38	37	Washington, DC*	43	27

Note: Quarters present (of 48) count flagged cells as present; "Clean" counts the unflagged quarters, which enter the access estimates. Metros contributing unflagged cells by year: 45 (2014), 48 (2015), 50–52 (2016–2023), 48 (2024), 44 (2025). *Washington, DC: bus operators only, no archived WMATA feed (Section 3.3).

Table A.2: Archive-integrity flag classes

Class	Detection rule	Cells
Partial outage	An operator present in the quarters before and after is missing from the quarter's archived feeds	73
Thin feed	The quarter's archived schedules contain far less service than the metro's other quarters	6
Archive scan	Access is more than 1.8 times the metro's median across quarters for up to four quarters, or less than 0.6 times it for three or more quarters, and the archived feeds show the cause (a feed included twice, or operators missing from the archive)	132
NTD divergence	For two or more consecutive years, the change in the network component and the change in the vehicle revenue miles the agencies reported to the NTD differ by 12 log-points or more, and the archived feeds confirm the archive as the cause (Fig. A.3)	72

Note: Cells are flagged metro-quarters per class; classes overlap, and some flagged cells fall outside the panel, so the distinct flagged total inside the panel is 136 of 2,330 metro-quarters (5.8%). Flagged cells are excluded from every access estimate and count as present for coverage (Section 3.3).

Table A.3: Representativeness: the 52-metro panel versus all U.S. metros

Metro-level median	Panel (52 metros)	All U.S. metros (392)
Population	1.6M	0.24M
Transit commute share (%)	1.7	0.7
Median household income ($000)	80.5	67.5
Car-free households (%)	6.0	6.0
Non-Hispanic White (%)	63.1	70.8
Non-Hispanic Black (%)	6.5	6.3
Hispanic (%)	13.4	8.1
Non-Hispanic Asian (%)	4.5	2.0

Note: Medians across metropolitan statistical areas, ACS 2018–2022 (tables B01003, B08301, B19013, B03002, B25044); New York excluded.

Table A.4: Jobs reachable by time budget

Panel A. Outside New York, 2023

Time budget	Transit	Car	Gap
15 min	1.9	164.1	86×
30 min	10.9	780.5	71×
45 min	37.6	1,433.7	38×
60 min	86.9	1,868.2	21×

Panel B. New York, 2023

Time budget	Transit	Car	Gap
15 min	11.3	527.0	47×
30 min	124.3	2,788.1	22×
45 min	410.3	5,013.2	12×
60 min	828.7	6,706.2	8×

Note: Population-weighted mean jobs reachable within each budget, in thousands, all jobs (C000); "Gap" = car/transit. Panel A is the 52-metro panel outside New York and Panel B is New York, both in 2023 with flagged cells excluded. Car times are free-flow (Section 3.1).

Table A.5: Correlation among access constructs

	Cumulative	Gravity	Logsum
Cumulative (45 min)	1.00	0.90 (0.90)	0.92 (0.91)
Gravity	0.90 (0.90)	1.00	1.00 (1.00)
Logsum	0.92 (0.91)	1.00 (1.00)	1.00

Note: Pearson correlations of block-group access scores in logs, from the routed 2019Q4 travel-time matrices, $n = 85{,}694$ block groups, 52 metros outside New York; pooled, with the median within-metro correlation in parentheses. Constructs: the paper's cumulative measure (jobs reachable within 45 minutes); a gravity measure with negative-exponential impedance, half-life 45 minutes ($\sum_j \text{jobs}_j \, e^{-(\ln 2/45)\, t_{ij}}$); and a destination-choice logsum ($\ln \sum_j \text{jobs}_j \, e^{-t_{ij}/45}$, a 45-minute decay constant, equivalent to a half-life of about 31 minutes).

Table A.6: Main statistics under block-group and metro clustering

		Block-group	Metro-clustered specification	
Statistic	Estimate	clustered	Estimate	Interval
Trajectory, 2025 (corrected)	+14.1	[+13.3, +14.8]	+14.9	[+6.1, +23.8]
at 2019Q4	+8.9	[+8.1, +9.7]	+5.2	[−9.2, +19.6]
Decomposition: job growth	+15.9	—	+15.0	[+11.8, +18.2]
job relocation	+5.7	[+5.1, +6.4]	+5.3	[+2.8, +7.8]
transit network	−6.4	[−7.0, −5.9]	−4.9	[−17.6, +7.9]
Growth, richest quartile	+30.9	[+28.8, +33.0]	+30.9	[+17.6, +45.7]
Growth, majority-Black (45 min)	−9.0	[−11.4, −6.5]	−9.0	[−18.2, +1.2]
Growth, majority-Black (60 min)	−12.5	[−15.0, −10.0]	−12.5	[−26.1, +3.5]
Within-city: poorest minus richest	−20.7	[−22.1, −19.2]	−20.7	[−26.8, −14.5]
majority-Black minus White	−10.7	[−13.0, −8.3]	−9.4	[−19.4, +0.6]

Note: Bracketed ranges are 95% confidence intervals. Trajectory and group growth: archive-entry-corrected within-block-group fixed-effects estimates (trajectory in log-points relative to the 2014 base, group growth in percent). Decomposition: 2025 vs. the 2014 average, log-points; job growth is the same for every block group in a metro, and only its metro-clustered interval is reported. Within-city contrasts: metro × 5-km cells absorbed, log-points. Point estimates are those reported in Sections 5.1–5.6; the metro-clustered specification's own point estimates are shown beside its intervals. Its trajectory rows compare 2025Q4 and 2019Q4 with 2014Q1, on block groups observed in at least 44 of the 48 quarters. Its decomposition rows use the components without the archive-entry correction, on the full 52-metro panel. Its majority-Black minus White contrast leaves out majority-Hispanic block groups. Its other rows use the same specification as the reported estimates. Metro clustering: ≈ 52 clusters (36 for the majority-Black growth rows, 45 for the within-city rows).

Table A.7: Reconciliation of the trajectory and decomposition estimands

	2025 endpoint
(1) Reported trajectory (52 metros, quarterly, archive-entry correction)	+14.1
(2) Same estimator on the every-year subset	+14.7
(3) Same estimator on its never-flagged metros	+16.1
(4) Decomposition observed (annual, jumps removed, same sample)	+15.2
Coverage sample step, (1)→(2)	+0.6
Archive-quality sample step, (2)→(3)	+1.4
Construction step, (3)→(4)	−0.9

Note: The trajectory estimand and the decomposition estimand, reconciled on a common basis (Section 5.2), 2025 vs. 2014 in log-points: the two sample steps are the every-year restriction and then the clean-archive (never-flagged) restriction; the construction step compares annual endpoint means with each block group's entry jumps removed against the quarterly fixed-effects form of the archive-entry correction.

Table A.8: Change in job access by transit across neighborhood groups, by time budget and classification year, 2014–2025

A. Time budget Group	30 min	45 min	60 min
Income quartile: poorest	+4.6	−0.3	−2.2
second	+14.2	+8.2	+4.5
third	+25.0	+18.5	+13.9
richest	+40.0	+30.9	+26.3
Distance to core: core quartile	+6.4	+3.8	+2.9
second	+10.7	+4.1	+1.9
third	+26.0	+15.6	+9.1
fringe quartile	+37.5	+31.8	+27.3
Majority-White	+21.2	+15.8	+12.6
Majority-Black	+0.5	−9.0	−12.5
Majority-Hispanic	+17.7	+11.2	+9.3

B. Classification year (45 minutes) Group	2018–2022	2009–2013
Income quartile: poorest	−0.3	−1.0
second	+8.2	+5.8
third	+18.5	+19.4
richest	+30.9	+30.7
Car-free households: all block groups	+2.5	+3.0
car-free share >10% ∩ poverty share >40%	+1.1	−1.2
car-free share >20% ∩ poverty share >50%	+2.9	−0.9

C. Within-city contrast, classification years as in Panel B (45 minutes, log-points)		
Poorest minus richest income quartile	−20.7 [−22.1, −19.2]	−21.7 [−23.2, −20.3]

Note: Panels A and B give the change in jobs reachable by transit from 2014 to 2025, in percent, from archive-entry-corrected within-block-group fixed-effects estimates for 2025 against the 2014 average, 52 metros outside New York, flagged cells excluded; point estimates. Panel A classifies block groups on the 2018–2022 ACS and weights them by population; its 45-minute column repeats Figs. 7 and 8. Panel B uses the 45-minute budget and classifies block groups on the 2018–2022 ACS or on the 2009–2013 ACS crosswalked to 2020 block groups; income rows are weighted by population, car-free rows by the car-free households of the classification vintage. Panel C is the difference between the poorest and richest income quartiles in the decade change of log access (2014–15 to 2024–25 window means, corrected for archive entry), estimated as in Table 3: metro × 5-km-distance-band cells absorbed, population weights, the 45 metros with both endpoint windows observed, log-points, robust 95% intervals.